

Algebra II

Հանրահաշիվ II

COURSE CODE	MTH-A2
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Typically Grade 10 or 11
DURATION	108-126 instructional hours (approx. 90 lessons)
PREREQUISITE	Algebra I (MTH-A1) and Geometry (MTH-GE), or placement by assessment.
LEADS TO	Precalculus (MTH-PC) — or AP Statistics (MTH-APS) as a parallel option

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Algebra II widens the universe of functions. Polynomial functions of higher degree, rational functions, radical functions, exponential and logarithmic functions, and the trigonometric functions all appear, and the complex numbers are introduced to complete the solution of polynomial equations. The unifying theme is that the same questions are asked of every family: what is its domain, what does its graph look like, how does it transform, how are its equations solved, and what does it model. Students who grasp this structural view find Precalculus and Calculus substantially easier.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Algebra II is often experienced as a long list of unrelated techniques. Վարժարան organizes it instead around a single recurring framework — the function family analysis — applied identically to each new family: parent function, transformations, domain and range, key features, inverse, equation solving, and modelling contexts. Students are given this framework in the first week and asked to complete it for every family thereafter. The result is that the eighth family is easier than the second, because the student knows what questions to ask. Teachers are also asked to insist on domain restrictions and extraneous-solution checking throughout, since these are the two habits that separate students who succeed in Calculus from those who do not.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Factors quadratics reliably and solves them by multiple methods	Polynomial and rational work in Units 2 and 3 depend entirely on factoring fluency.
Interprets function notation and transformations of a parent function	The organizing framework of the course; if absent, Unit 1 must be extended.
Simplifies radical and rational expressions	Prerequisite for Units 3 and 4; a common gap after a geometry year.
Solves systems of linear equations	Extended in this course to nonlinear systems and to partial fractions.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Perform arithmetic with complex numbers and solve quadratic equations with complex solutions.
- 2 Analyze functions using key features, transformations, compositions, and inverses.
- 3 Understand polynomial functions: end behaviour, zeros, multiplicity, and the relationship between factors and roots.
- 4 Apply the Remainder and Factor theorems and the Fundamental Theorem of Algebra.
- 5 Rewrite rational expressions and solve rational equations, identifying extraneous solutions.
- 6 Analyze rational functions including asymptotes, holes, and end behaviour.
- 7 Solve radical equations and analyze radical functions, with attention to domain.
- 8 Understand logarithms as inverses of exponentials and apply the properties of logarithms.
- 9 Solve exponential and logarithmic equations and model with them, including natural base contexts.
- 10 Extend trigonometric functions to all real numbers using the unit circle and radian measure.
- 11 Model periodic phenomena with trigonometric functions and prove and apply basic identities.
- 12 Work with arithmetic and geometric sequences and series, including infinite geometric series.
- 13 Summarize and interpret data, understand normal distributions, and evaluate reports based on data.
- 14 Make inferences and justify conclusions from sample surveys, experiments, and observational studies.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 12-13 LESSONS

Functions, Transformations, and Complex Numbers

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TOPICS COVERED

- Review of function notation, domain, and range
- The parent function catalogue: linear, quadratic, cubic, absolute value, square root, reciprocal, exponential
- Vertical and horizontal translations, reflections, stretches, and compressions
- The order in which transformations are applied and why it matters
- Writing a function from a described transformation and vice versa
- Even and odd functions and their symmetry
- Piecewise-defined functions and their graphs
- Composition of functions and the domain of a composition
- Inverse functions, the horizontal line test, and finding an inverse algebraically
- The graphical relationship between a function and its inverse
- Imaginary and complex numbers; the need for i
- Arithmetic with complex numbers; conjugates and division
- Solving quadratic equations with complex solutions
- The complex plane and the modulus

LEARNING OBJECTIVES

- 1 Describe a transformation sequence from an equation and produce an equation from a description.
- 2 Determine the domain of a composition and explain why it may be smaller than expected.
- 3 Find an inverse function algebraically and verify it by composition.
- 4 Perform all four operations with complex numbers and solve quadratics with complex roots.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Describes and applies transformations correctly in 9 of 10 trials.
- Finds and verifies inverse functions in 9 of 10 trials.
- Computes with complex numbers at 92% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Horizontal transformations are applied in the intuitive rather than the actual direction: $f(x - 3)$ shifted left.	The input must reach the same value, so the graph shifts right. Build a table of values and compare — the table settles it in a way that assertion cannot.

MISCONCEPTION	RECOMMENDED RESPONSE
$f^{-1}(x)$ is read as a reciprocal.	It denotes the inverse function. Contrast with $1/f(x)$ explicitly in the same lesson; this confusion is common and consequential.
The domain of a composition is taken as the domain of the outer function alone.	The inner output must be a legal input to the outer function. Work through an example where the composition's domain is strictly smaller.

TEACHING NOTE

The function family framework is introduced here and used for the rest of the course. Give the student a written template in the first week and require it to be completed for every family that follows; the cumulative effect over the year is substantial.

UNIT 2 · 12-13 LESSONS**Polynomial Functions**

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TOPICS COVERED

- Polynomial vocabulary and standard form
- End behaviour determined by degree and leading coefficient
- Zeros, factors, and x-intercepts
- Multiplicity and its effect on the graph at a zero
- Sketching a polynomial from its factored form
- Polynomial long division and synthetic division
- The Remainder Theorem and the Factor Theorem
- The Rational Root Theorem and systematic root finding
- The Fundamental Theorem of Algebra and the count of roots with multiplicity
- Complex roots occurring in conjugate pairs
- Writing a polynomial from its roots
- Solving polynomial equations and inequalities
- Sum and difference of cubes; factoring by grouping in higher degree
- Polynomial identities and the binomial theorem with Pascal's triangle
- Modelling with polynomial functions

LEARNING OBJECTIVES

- 1 Predict end behaviour and sketch a polynomial from its factored form, including multiplicity effects.
- 2 Use synthetic division with the Rational Root Theorem to factor a higher-degree polynomial completely.
- 3 State and apply the Fundamental Theorem of Algebra, accounting for complex and repeated roots.
- 4 Construct a polynomial with specified roots and leading coefficient.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Sketches polynomial graphs with correct end behaviour and multiplicity behaviour in 9 of 10 trials.
- Factors higher-degree polynomials completely at 88% accuracy.
- Solves polynomial inequalities correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Every zero is assumed to be a crossing point regardless of multiplicity.	Even multiplicity touches; odd multiplicity crosses. Graph $(x-1)^2(x+2)$ and observe.
The Fundamental Theorem is taken to guarantee n distinct real roots.	It guarantees n roots in the complex numbers, counted with multiplicity. Be precise in the statement every time.
Polynomial inequalities are solved by treating them as equations and ignoring sign analysis.	Use a sign chart with the zeros as boundaries. Test each interval; do not reason from the equation alone.

TEACHING NOTE

The sign chart introduced here is used again for rational inequalities, for domain analysis, and throughout Calculus. Teach it as a general tool rather than as a polynomial-specific trick.

UNIT 3 • 12-13 LESSONS

Rational and Radical Functions

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TOPICS COVERED

- Simplifying rational expressions and identifying restricted values
- Multiplying, dividing, adding, and subtracting rational expressions
- Complex fractions
- Solving rational equations and identifying extraneous solutions
- Rational inequalities and sign analysis
- The reciprocal parent function and its transformations
- Vertical asymptotes, holes, and the distinction between them
- Horizontal and slant asymptotes from degree comparison
- Sketching a rational function completely
- Direct, inverse, joint, and combined variation
- Simplifying radical expressions and rationalizing denominators
- Rational exponents and their equivalence to radicals
- Solving radical equations and checking for extraneous solutions
- Radical functions, their domains, and their graphs
- Modelling with rational and radical functions

LEARNING OBJECTIVES

- 1 Simplify a rational expression and state all restricted values, including those lost in simplification.
- 2 Solve a rational or radical equation and identify every extraneous solution with a reason.
- 3 Determine all asymptotes and holes and sketch a complete rational function graph.
- 4 Model a real situation with an inverse or combined variation relationship.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Simplifies and operates on rational expressions at 88% accuracy with restrictions stated.
- Identifies extraneous solutions correctly in 9 of 10 trials.
- Sketches rational functions with all features correct in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Terms are cancelled across addition: $(x + 3)/3$ simplified to x.	Only common <i>factors</i> cancel. Substitute a number into both forms to demonstrate the inequality; this is far more effective than restating the rule.
Extraneous solutions are treated as arithmetic mistakes rather than as a structural consequence.	Squaring and multiplying by a variable expression are not reversible operations. Explain the mechanism, then require the check permanently.
Every zero of the denominator produces a vertical asymptote.	If the factor cancels, it produces a hole. Factor both numerator and denominator fully before deciding.

TEACHING NOTE

Requiring domain restrictions to be stated at every step — not only in the final answer — is the habit that makes rational functions safe. It also transfers directly to Calculus, where domain errors are the most common source of invalid conclusions.

UNIT 4 • 12-13 LESSONS**Exponential and Logarithmic Functions**

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TOPICS COVERED

- Exponential functions, growth and decay, and transformations
- The number e and continuous growth
- Logarithms defined as inverses of exponentials
- Common and natural logarithms
- Converting between exponential and logarithmic form
- Graphs of logarithmic functions, their domains, and asymptotes
- Properties of logarithms: product, quotient, and power
- The change of base formula
- Expanding and condensing logarithmic expressions
- Solving exponential equations using logarithms
- Solving logarithmic equations and checking the domain
- Compound and continuously compounded interest
- Exponential growth and decay models; half-life and doubling time
- Newton's law of cooling and logistic growth introduced informally
- Fitting an exponential model to data using logarithms
- Logarithmic scales: pH, decibels, and the Richter scale

LEARNING OBJECTIVES

- 1 Convert fluently between exponential and logarithmic form and explain the inverse relationship.
- 2 Apply the logarithm properties to expand, condense, and solve.
- 3 Solve exponential and logarithmic equations, rejecting solutions outside the domain.
- 4 Build and interpret an exponential or logarithmic model of a real situation.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Applies logarithm properties correctly at 90% accuracy.
- Solves exponential and logarithmic equations at 88% accuracy with domain checking.
- Interprets model parameters in context with units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
$\log(a + b)$ is expanded as $\log a + \log b$.	The product property applies to products, not sums. Test numerically with a calculator; the counterexample is immediate and permanent.
Negative or zero solutions to logarithmic equations are accepted.	The argument of a logarithm must be positive. Domain checking is mandatory, not optional.
The logarithm is treated as an operation applied to one side of an equation only.	It applies to both sides, like any function. Write it on both sides explicitly.

TEACHING NOTE

The inverse relationship should be the organizing idea of the whole unit, not a remark in the first lesson. Every logarithm property has an exponential counterpart, and presenting them in pairs roughly halves the memorization load.

UNIT 5 • 12-13 LESSONS

Trigonometric Functions

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TOPICS COVERED

- Angles in standard position; coterminal and reference angles
- Radian measure and conversion
- Arc length and sector area in radians
- The unit circle and the extension of sine and cosine to all real numbers
- Exact values at the special angles
- The signs of the functions by quadrant
- Graphs of sine and cosine; amplitude, period, phase shift, and vertical shift
- Graphs of tangent, cotangent, secant, and cosecant
- Writing a trigonometric function from a described periodic situation
- Modelling periodic phenomena: tides, daylight hours, temperature cycles
- The Pythagorean identity and its derivation from the unit circle
- Reciprocal, quotient, and even-odd identities
- Sum and difference formulas
- Double-angle formulas
- Solving basic trigonometric equations over a stated interval
- Inverse trigonometric functions and their restricted domains

LEARNING OBJECTIVES

- 1 Use the unit circle to determine exact values and to justify the Pythagorean identity.
- 2 Graph any sinusoid from its equation and write an equation from a graph or a described situation.
- 3 Solve a trigonometric equation over a specified interval, finding all solutions.
- 4 Model a periodic real-world phenomenon and interpret each parameter.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces exact values at special angles in 9 of 10 trials without a calculator.
- Graphs and writes sinusoids at 90% accuracy including all four parameters.
- Solves trigonometric equations over an interval, finding all solutions, in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The calculator is left in the wrong angle mode.	Check the mode at the start of every lesson and before every assessment. It is trivial and it prevents a great deal of confusion.
Only one solution is reported for a trigonometric equation over an interval.	Periodic functions repeat. Sketch the graph and count the intersections before solving algebraically.

MISCONCEPTION	RECOMMENDED RESPONSE
The phase shift is read directly from the equation without factoring out the horizontal coefficient.	In $y = \sin(2x - \pi)$, the shift is $\pi/2$, not π . Factor first, always.

TEACHING NOTE

The unit circle should be constructed by the student, not handed to them. A student who has built it — including reasoning through the special triangles — retains it; a student who has memorized a printed chart does not.

UNIT 6 • 11-12 LESSONS**Sequences, Series, and Statistical Inference**

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TOPICS COVERED

- Arithmetic and geometric sequences; explicit and recursive forms
- Arithmetic series and the derivation of the sum formula
- Geometric series and the derivation of the sum formula
- Infinite geometric series and the condition for convergence
- Sigma notation
- Applications: annuities, loan amortization, and repeating decimals as series
- Sampling: random samples, surveys, experiments, and observational studies
- What each study design can and cannot establish
- Sources of bias and how randomization addresses them
- Normal distributions and the empirical rule
- Using the normal distribution to estimate population percentages
- Sampling variability and the idea of a margin of error
- Simulation to model random behaviour and to assess significance informally
- Comparing two treatments and deciding whether a difference is significant
- Evaluating a report based on data and identifying misleading presentations

LEARNING OBJECTIVES

- 1 Derive and apply the arithmetic and geometric series sum formulas.
- 2 Determine whether an infinite geometric series converges and find its sum.
- 3 Distinguish what an experiment can establish from what an observational study can.
- 4 Use a simulation to judge whether an observed difference is likely to be due to chance.
- 5 Critique a data-based claim in a news report and identify its weaknesses.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Works with sequences and series at 90% accuracy including sigma notation.
- Correctly classifies study designs and their valid conclusions in 9 of 10 trials.
- Produces a defensible critique of a data-based claim in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Every infinite series is assumed to converge, or none are.	A geometric series converges exactly when $ r < 1$. Compute partial sums for $r = 0.5$ and $r = 2$ and compare the behaviour.
An observational study is used to support a causal claim.	Only a randomized controlled experiment supports causation. Require the student to state which design was used before evaluating any conclusion.
The empirical rule is applied to distributions that are not approximately normal.	Check the shape first. The rule is a property of normal distributions, not of data in general.

TEACHING NOTE

This unit pairs two topics that are usually separate, and the pairing is deliberate: infinite series and statistical inference are both about drawing a firm conclusion from an incomplete process. For students continuing to AP Statistics, the second half is a direct on-ramp.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Algebra I retention with emphasis on factoring, functions, and radicals.
Unit Check	End of each unit · 55 min	Unit mastery criteria plus a completed function family analysis for the family studied.
Function Family Portfolio	Continuous	The student's completed analysis template for every family; the structural record of the course.
Fluency Probe	Every 3rd lesson · 5 min	Factoring, exponent and logarithm properties, and unit circle values.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Precalculus placement and readiness for the AP pathway.

Վարժապան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard Year	2 lessons/week, 60–75 min	Full course in approximately 45 weeks.
Extended	1 lesson/week, 75 min	Full course across two academic years.
Accelerated	3 lessons/week	Full course in approximately 30 weeks; for students targeting AP Calculus in Grade 11.
Support / Co-Seated	2 lessons/week alongside school	Paced to the student's school syllabus.
Summer Program	4 lessons/week, 8 weeks	Units 4 and 5 — logarithms and trigonometry — the most common summer request at this level.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Sixty minutes per lesson: twelve to eighteen problems in three parts — current content, cumulative review spanning all previous units, and one extended or modelling problem. Domain restrictions must be stated wherever they exist, and solutions to rational, radical, and logarithmic equations must be checked with the check shown; unchecked solutions are marked incomplete regardless of correctness. The function family analysis template is updated as each new family is introduced and is submitted with the unit check.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Analyzes any function family using a consistent structural framework.
- ◆ Composes functions and finds inverses, with correct domain reasoning.
- ◆ Computes with complex numbers and solves quadratics with complex roots.
- ◆ Analyzes polynomial functions using end behaviour, zeros, and multiplicity.
- ◆ Applies the Remainder, Factor, and Rational Root theorems and the Fundamental Theorem of Algebra.
- ◆ Works with rational expressions and identifies asymptotes and holes.
- ◆ Solves rational and radical equations and identifies extraneous solutions.
- ◆ Understands logarithms as inverses and applies their properties.
- ◆ Solves and models with exponential and logarithmic functions.
- ◆ Uses the unit circle and radian measure to extend trigonometry to all real numbers.
- ◆ Graphs and models periodic phenomena with trigonometric functions.

- ◆ Works with sequences and series, including infinite geometric series.
- ◆ Evaluates study designs and critiques data-based claims.
- ◆ Completed Algebra II — ready for Precalculus.

MATERIALS AND RESOURCES

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- Graphing calculator (TI-84 family or equivalent) — required throughout.
- Dynamic graphing software for transformation exploration.
- Student-constructed unit circle reference, built rather than issued.
- Function family analysis template (Վարժարան), completed across the year.
- Վարժարան Algebra II problem bank — approximately 2,200 items with a 200-item extraneous-solution sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.